

**A framework for a systems design approach to complex societal problems: analysis and synthesis**

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### Abstract

The paper start with a deep elaboration of the article “*A framework for a systems design approach to complex societal problems*” from researchers at Delft university Jairo da Costa Junior, Jan Carel Diehl and Dirk Snelders published in 2019. After a review of the topics, connection was made with my personal knowledge about design methods and processes acquired from Alta Scuola Politecnica classes during my masters’ studies. The framework developed in the paper turns out to be very concise and strong to complement design tools and methods, as desired by the author. Subsequently, a sport case of designing a basketball practice was analyzed, and the relative framework was developed, showing the potentialities for such techniques discussed by the authors of the paper. Moreover, the conclusion provides some personal suggestions about possible research developments on the topic.

### **A framework of system design approach to complex societal problems**

In the assigned paper the authors start by characterizing societal problems as inherently complex, intertwining social, technical and political dimensions.

They argue that a paradigm shift is needed by designers to handle such complexity, pointing to the fusion of Systems Theory and Design as the possible solution.

While Design is about changing existing situations into preferred ones, understanding mechanisms and create; Systems theory is a conceptual framework assuming the relatedness of the component parts of a system with each other and with other systems (holistic view). The collaboration between the two fields will be called System Design.

#### System Thinking

Traditional scientific methods rely on reductionism and analytical thinking; however, such methods cannot anticipate side effects and struggle with dependencies. Holism is needed to handle the intertwined complexity.

The way of thinking based on these assumptions is called *System thinking*; and such can bring numerous advantages like better complexity handling and consideration of many stakeholders' perspectives.

Different ways of thinking lead to different approaches on which we base our design process. Thinkings are classified based on objectives and methodologies on Hard, Soft and Critical systems thinking.

*Hard systems thinking* is used when the objective is maximization of a chosen parameter. Hard Systems Methodologies (HSM) include operational research,

system analysis (SE), systems engineering and aim to predict and control systems output.

HSM assume that starting conditions are known, the problem is well defined and that real world problems can be modelled, and therefore its behaviour can be simulated.

The side effect is that choosing one problem solution, neglects all the other possibilities creating possible lacks.

*Soft System thinking* adopts subjectivist perspective to put stakeholders' interest at the core of the solution. Soft Systems Methodologies (SSM) seek to understand individual interpretations of the world to create a shared consciousness and mutual agreement about the action to be taken.

SSM treats stakeholders as co-creators of value and requires a collaboration. This brings as positive effects an increased commitment and ask for respect of world views. The danger is to listen only one perspective and exclude the others. However, this allows designers to work with uncertainty and unpredictability, shifting from a concept of optimizing to a concept of learning.

*Critical thinking systems* are starting from SSM idea, but they implement ways to delay possible coercive relationship effects in open debates, therefore Critical Systems Methods (CSM) gives boundary questions (who benefits, who bears costs, who decides) to force those in power to justify choices and open the process to those affected.

#### Design approaches

While systems thinking provides a base for synthesising knowledge, design methods bring innovative inquiry and creative actions. System design refers to the model

through which one, as a designer, can see the world. Four representative approaches are highlighted:

*Whole systems design* aims to provide a collective response to complex problems. Based on a holistic view, it supposes an active participation of all the stakeholders to the design process, where solutions must not be imposed, communication and comprehension must dominate.

*Systems-oriented design* (SOD) has the goal of making the system able to address its functions and be sustainable. By considering a hierarchy structure and boundaries among the system, it is a skill-based approach. An example is GIGA-mapping.

*Product-service systems* (PSS) are based on a fundamental assumption: human needs can be fulfilled through products.

*Design for development* (DfD) expect users to be empowered to act; solutions try to improve capabilities in stakeholders; they're mindful of sustainability (of future generations) and of power relations.

### Conceptual framework

To guide designers in adopting systems thinking, the authors propose a framework of five clusters: mindset, methodology set, knowledge set, skill set and tool set.

*System mindset* consists of three pillars to complement traditional design approaches:

- I. Find the assumption limitations
- II. Apply System thinking, knowing where the outcome result can be achieved
- III. Adopt holism, multi-stakeholder views, and complexity tolerance.

*System methodology*: one must be aware of the paradigm through which the stakeholder is observing the socio-technical

system and the paradigm used in the designing method. Information collected on one side may not be adapted to develop the desired solution.

*System knowledge*: Creation and collection of knowledge play a fundamental role in the design process. However knowledge can have its limitations, and it is strongly influenced by the collectors' human interests, which can be technical, practical or emancipatory interests.

In our three methods the type of knowledge required is different. For HSM knowledge is instrumental, predominating a quantitative form aiming to create a *causal explanation*. In SSM there is a practical interest in *understanding* all stakeholders' perspective and create a common knowledge, qualitative data is privileged. Lastly, for CSM both qualitative and quantitative data are needed but the fundamental role is played by *reflection*, aiming to understand power structures and protect minorities from such.

*System skills*: five skills are highlighted in the test as necessary; multilevel perspective, complexity handling, adaptability, ability to handle different stakeholders' points of view, the ability to work in multidisciplinary teams. For example, in the case of an energy solution for low-income households in rural areas designers will aim to create solutions at micro (new products and patterns), meso (among utilities and non-governmental organizations) and macro (pushing forward the adoption of new policies) level, reaching more stability and more satisfaction among different stakeholders. Doing so they will encounter the need of working in a team with different competencies and backgrounds.

*System tools*: must be used to facilitate collaborative inquiring, visualize, reason

and simulate. Systems methodologies provide tools to see, understand and find action such as pictures, conceptual models. Tools are system-way-of-thinking-dependent.

### **Personal elaboration**

#### The role of knowledge

The methodologies seen in the paper are essential and common ones. The “Smart Shopping Cart” exercise (studied during DM&P classes) fits an SSM: everybody in class was giving out loud his/her opinion to conquer knowledge and find a new solution, the system was free from coercive relationships of power and interests. The paper stresses careful how the collection of knowledge is crucial to the kind of system, and different systems require different sets of knowledge. Indeed, in Concept-Knowledge theory everything starts from the knowledge acquired, which is clustered into concepts and then expanded through negation. Knowledge directly shapes concepts, and one could argue that different set of knowledge produce different outcomes.

#### Contextuality

Design innovation arises from the dynamic interplay between research and creativity. One can recall the importance of research defined as the “formal work undertaken systematically to increase the stock of knowledge”. The paper gives us a guide on how to perform the research activity that (not always) precedes creative processes; when to search for qualitative and/or quantitative data, depending on the type of the system.

On the other hand, context is important also for creativity and different methods may be applied in different contexts.

At the same time, it is highlighted how different context bring to different solution and processes, in fact knowledge is need dependent, objectives-dependent, resistance to change is context dependent, while solution selection process is depending on many subjective, political and constraining factors.

The context in which the design process happens appear to be fundamental for its success.

#### Stakeholders' perspective

Stakeholders' thoughts and impression play a fundamental role in SSM and CSM. However, their role extends also to post-creation steps, with the resistance to change. Naturally people tend to resist to change, and this is fostering a barrier between the ideation and the actual implementation of designed products or services.

SSM and CSM are professing the need of an open (and fair) debate among stakeholders, innovation cannot come from the interests of only one powerful member. This message is core of the design strategies to accelerate technology: innovation takes place when design balance benefits and resistance across multiple stakeholders, and an open discussion can bring closer to such balance. *Systems perspective* is highly required as in DfD.

Moreover, because of what mentioned above, the formulation of stakeholders' profiles remains necessary for a proper conceptualization of the market situation and to better visualize corresponding needs and requirements.

Moreover, one could argue that stakeholders' perspective can shape the system category through their needs. Human-Based needs will be prominent in Soft Systems, in Critical Systems also

regulatory needs will appear and technical or business needs will be more important in Hard Systems.

### **Personal experience**

I've been playing basketball for 17 years, and by delving deeply into these topics I immediately recognize the importance of design in the sport field. I will talk more specifically about basketball, however with strong conviction that most of the concepts will be applicable also in other team sport cases.

The role of a basketball coach is fundamental, and he/she can be considered a designer. The coach is not only choosing tactics during games but designing practices every week, all season long.

The team can be considered as a Soft System: every team member, the sport club and coaches are all stakeholders with the same interest in winning and, usually, there is no conflict of interests.

Sports are increasingly being described as complex systems and cannot be described by isolated components (Scott Mclean, 2025).

According to my personal experience there is a huge lack of competence among basketball agonistic coaches, even at "good" levels. Researchers agree that training should be continuous, more specialized, and comprehensive, focusing on understanding and constructing knowledge that supports the professional growth of basketballers (Alemany-Iturriaga, 2025); while nowadays it is not enough. This leaves a lot of space for innovation and possibly big results from System Design applications; indeed, the existing literature is not exhaustive.

Design approach is *System Oriented Design*: a hierarchic structure is needed between coaches and players, and among

players themselves, even though everybody should have the possibility of speaking out. In such a complex system, the authors' framework turns out to be a very powerful tool to guide Basketball coaches' design. Some general acknowledgements can be considered for the starting framework:

*System mindset*: the assumptions limitations to apply are that not every player will react physically and psychologically in the same way at the same practice, personalization is necessary for optimal results.

*System methodology*: when asking for players' opinions one must pay attention on personal interests.

*System knowledge*: knowledge should be both quantitative and qualitative; academic research will show most effective training plans, however players' emotions are crucial to maintain a good and motivating environment.

*System skills*: a very good practice can be not effective for many reasons, such as team cohesion and engagement, therefore a good coach must pay attention to a multitude of factors (multilevel approach). Coaches must handle an entire team of people with motivational speeches and authority (complexity handling) to ensure players will follow the hierarchy; but at the same time a coach must adapt the games on characteristics of the team players such as good shooting abilities, physical strength and athleticism (adaptability).

*System tools*: game video rewatching is a must for every coach to better understand all the details and gain more knowledge, however generic designing tool for ideating new exercises and practices could be implemented to find new ideas and solutions.

### **Comments**

I found this article very interesting, and I would like to cite my deep interest for politics and history. I noticed that international politics can be considered a complex Critical System, with different stakeholders and coercive relationships among different states. Therefore, I would like to see if it is possible to design a sustainable system for international peace while at the same time considering many aspects as human rights, economic development and people opinions. Is it possible to satisfy the desire of peace for all stakeholders? Or should one stop caring about human rights in further regions if there is the desire for a certain degree of wealth and peace in his/her own country? Where should be the limit of so?

Is peace possible? Is economical equity possible? I would be very interested in considering some international relations design models to such answers, and to find out the results. What mechanisms should international organization develop to maintain authority and peace?

### **Conclusion**

The paper highlights some very interesting facts about System Thinking and Design, providing a powerful bridge among the two fields to guide designers in their role. Authors properly developed a powerful tool that could be applied in many different cases to guide designers through their process and increase success of innovations.

## References

- Alemany-Iturriaga, J. (2025). Methodology and content for the design of basketball coach education programs: a systematic review.
- Scott Mclean, M. N. (2025). *Systems Thinking Methods in Sport: Practical Guidance and Case Study Applications*.